

Combinatorics Basics

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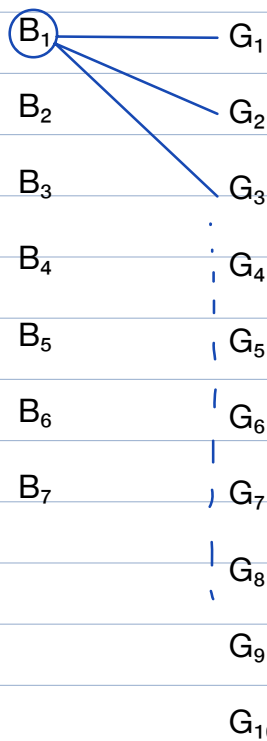
Multiplication / Addition Rule

< Question > : Given 10 Girls and 7 Boys. How many different pairs?

Pair $\rightarrow$ 1 Girl & 1 Boy

Boys

Girls



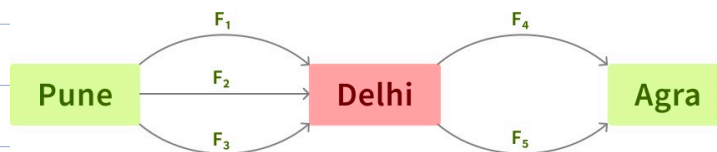
$$7 * 10 = 70 \text{ pairs}$$

Annotations: 7 (1 Boy), * (B), 10 (1 Girl)

And $\rightarrow$ *



Example-1 :



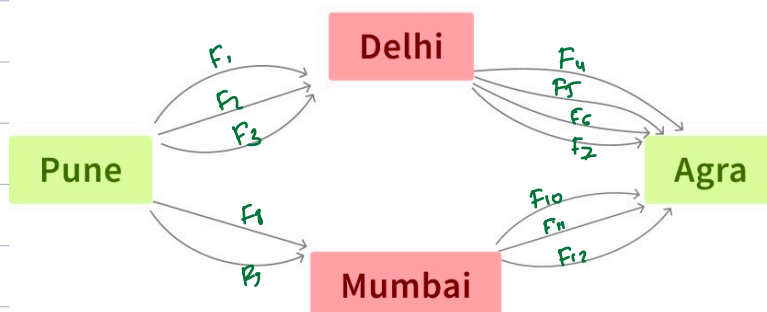
Number of ways to reach Agra from Pune via Delhi

$$\begin{array}{c}
 P \longrightarrow D \quad \& \quad D \longrightarrow A \\
 \underbrace{\hspace{1cm}} \quad \underbrace{\hspace{1cm}} \\
 3 \qquad \qquad \qquad 2
 \end{array}
 \qquad
 \begin{array}{ccc}
 F_1 F_4 & F_2 F_4 & F_3 F_4 \\
 F_1 F_5 & F_2 F_5 & F_3 F_5
 \end{array}$$

$$3 * 2$$

Example-2 :

Number of ways of reaching Agra from Pune ?



$$\begin{array}{c}
 P \longrightarrow D \quad \& \quad D \longrightarrow A \\
 3 \qquad \qquad \qquad * \qquad \qquad \qquad 4 \\
 \qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad = 12
 \end{array}$$

OR $\longrightarrow$ +

$$12 + 6 = 18$$

$$\begin{array}{c}
 P \longrightarrow M \quad \& \quad M \longrightarrow A \\
 2 \qquad \qquad \qquad * \qquad \qquad \qquad 3 \\
 \qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad = 6
 \end{array}$$

Zomato, features an exciting option for its users - meal combos. Each combo includes one main course, one *dessert*, and one *beverage*, offering a complete dining experience from various restaurants. Zomato believes that a greater variety of combos can significantly enhance customer satisfaction.

Problem

You're tasked with helping **Zomato** identify which restaurant offers the most variety in its meal combos. You're provided with a list, shaped like a grid or a 2D matrix **A**, where each row corresponds to a different restaurant's offerings.

Each row is divided into three parts:

1. $A[i][0]$ tells you the number of main courses,
2. $A[i][1]$ the number of desserts, and
3. $A[i][2]$ the number of beverages a restaurant offers.

Your challenge is to analyze this data and pinpoint which restaurant gives its customers the most options to mix and match their meal combo.

constraint $\rightarrow$ +ve

Eg $\rightarrow$

[

output

[3 2 2]

[4 3 3]

Max ($A[i][0]$) *

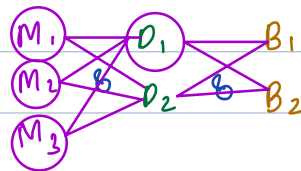
[1 1 1]

$A[i][1]$ *

]

$A[i][2]$

R₁





Permutations

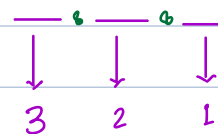
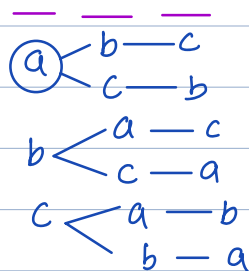
No. of arrangements

Order matters
 $(i, j) \neq (j, i)$

< Question-1 > : Given 3 distinct characters. In how many ways, we can arrange them?

S = "a b c"

$\begin{matrix} a & b & c \\ a & c & b \end{matrix}$
 $\begin{matrix} b & a & c \\ b & c & a \end{matrix}$
 $\begin{matrix} c & a & b \\ c & b & a \end{matrix}$
 } 6 arrangements.

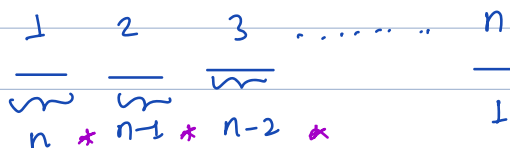


$$3 * 2 * 1 = 3! = \underline{\underline{6}}$$

< Question-2 > : In how many ways, you can arrange 4 distinct characters?

$$4! = 24$$

How many ways n distinct characters can be arranged



$n!$



< Question > : Given N distinct characters, in how many ways can we arrange r characters?

s = "date"

$$\underline{4} \quad \underline{3}$$

$$4 * 3 = 12$$

$$n - r - 1$$

1	2	3	4	5	...	r
—	—	—	—	—	...	—
N	N-1	N-2	N-3	N-4	...	N-(r-1)
						N-r+1

$$\Rightarrow \frac{N * N-1 * N-2 * N-3 * \dots * N-r+1 * N-r \dots 1}{N-r \dots 1}$$

$$\Rightarrow \frac{N!}{(N-r)!} \Rightarrow {}^N P_r$$

s = "date"	N = 4	$\frac{4!}{(4-2)!} = \frac{24}{2} = \underline{\underline{12}}$
	r = 2	



Combinations

$N C_r$ → selecting r objects from N objects

no. of ways to select

order

doesn't matter

$$(i, j) = (j, i)$$

< Question > : In how many ways can we select 3 players from a pool of 4 players?

[P1 P2 P3 P4]

$P_1 P_2 P_3 = P_1 P_3 P_2$

4 ways

$P_1 P_3 P_4$

$P_1 P_2 P_4$

$P_2 P_3 P_4$

Formula

$${}^4C_3 = \frac{4!}{3!(4-3)!} = \frac{4!}{3!} = 4$$

< Question > : Number of ways to arrange the players in 3 slots

Given 4 players → [P1 P2 P3 P4]

$\begin{matrix} - & - & - \\ 4 & 3 & 2 \end{matrix}$

$$= 24$$

$\left\{ \begin{matrix} P_1 P_2 P_3 \\ P_1 P_3 P_2 \\ P_2 P_3 P_1 \\ P_2 P_1 P_3 \\ P_3 P_1 P_2 \\ P_3 P_2 P_1 \end{matrix} \right\}$
 $\left\{ \begin{matrix} P_1 P_3 P_4 \\ P_1 P_2 P_4 \\ P_2 P_3 P_4 \end{matrix} \right\}$
 $\left\{ \begin{matrix} P_1 P_2 P_4 \\ P_2 P_3 P_4 \end{matrix} \right\}$

$$24$$

For every selection 3! arrangements are possible.

arrangement ${}^N P_r = \# \text{selecting } {}^N C_r * r!$

$${}^4 P_3 = {}^4 C_3 * 3!$$

$$\frac{{}^4 P_3}{3!} = {}^4 C_3$$

$$\frac{4 * 3 * 2}{3 * 2}$$

$$= 4$$

$$= {}^4 C_3$$

No. of ways to select r items from N items

$\Rightarrow$ NO. of arrangements ${}^N P_r =$ no. of selection ${}^N C_r * \text{no. of ways to arrange } r \text{ objects}$

$${}^N P_r = {}^N C_r * r!$$

$$\Rightarrow \frac{N!}{(N-r)!} = {}^N C_r * r!$$

$\Rightarrow$

$${}^N C_r = \frac{N!}{(N-r)! * r!}$$



Properties

1. ${}^N C_0 \longrightarrow$ select 0 items from N items = 1

$$\frac{N!}{0! * (N-0)!} = \frac{\cancel{N!}}{\cancel{N!} \cdot 0!} = 1$$

2. ${}^N C_N \longrightarrow$ select N items from N items = 1

P_1, P_2, P_3 No of ways to form a team of size 3

$${}^N C_N = \frac{N!}{N! * (N-N)!} = \frac{\cancel{N!}}{\cancel{N!} * 0!} = 1$$

3. ${}^N C_{N-r}$

$$\frac{N!}{(N-r)! * (N-(N-r))!} = \frac{N!}{(N-r)! * (r)!}$$

$${}^N C_{N-r} = {}^N C_r$$

Pranjal

C_1

C_2

C_5

C_3

C_4

Kumar

$C_1 C_2 C_3 C_4 C_5$

Rishav

$C_2 C_3 C_4 C_5$

4

$C_1 C_3 C_4 C_5$

$${}^5 C_1 == {}^5 C_4$$

$C_1 C_2 C_4 C_5$



Break

10:39

Special Property

< Question > : Given N distinct elements, select r distinct elements.

 $N C_r$ 1 2 nth element

select

or

not select

$${}^{N-1}C_{r-1}$$

$${}^{N-1}C_r$$

$$n C_r = {}^{n-1}C_{r-1} + {}^{n-1}C_r$$

$$5 C_3 = 4 C_2 + 4 C_3$$

$$\frac{5!}{3! 2!} = \frac{4!}{2! 2!} + 4 C_1$$

$$\underline{\underline{10}} = \underline{\underline{6}} + 4 = \underline{\underline{10}}$$

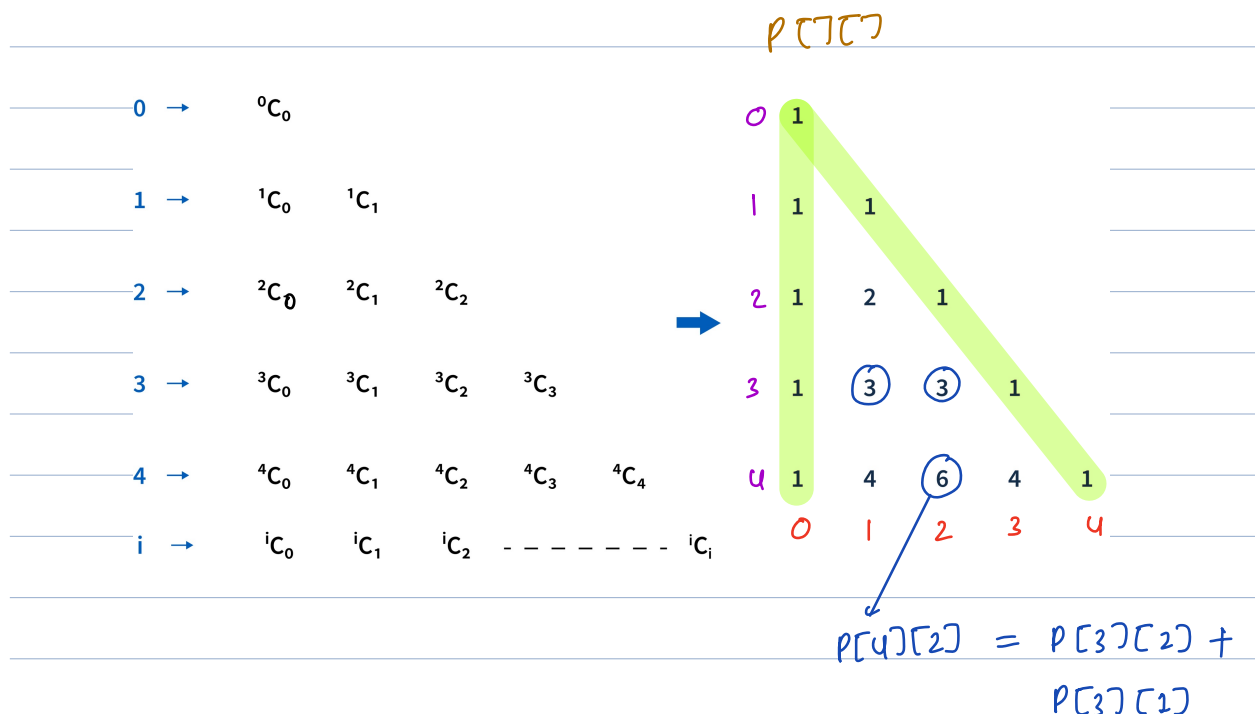
 $P_1 P_2 P_3 P_4$

Team = 2



Pascal Triangle

- Generate the Pascal's triangle for given N



Bruteforce approach

For every $n \in \{\text{row}\}$ and $c \in \{\text{col}\}$
 compute nC_c

```

for n → 0 to N {
  for c → 0 to n {
    compute ( ${}^nC_c$ )
  }
}

```

The complexity of the above code is $O(N^3)$.



</> Code

 $P[N][N]$ TC: $O(N^2)$ SC: $O(1)$ for $x \rightarrow 0$ to N {for $\rightarrow 0$ to x {

print

if ($c == 0$ || $c == x$) {SC: $O(N^2)$ $P[x][c] = 1$

} else {

 $P[x][c] = P[x-1][c] +$
 $P[x-1][c-1]$

}

}

}



Nth Column Title

{MIFT}

- Find the Nth column title

Given N $\longrightarrow$ column title

N = 1 2 3 26 27 28 50 51 52 53 54 ---
 A B C --- Z AA AB ----- AX AY AZ BA BB --- BZ

N = 30 $\rightarrow$ "AD"A $\rightarrow$ 1 -1 0N = 50 $\rightarrow$ "AX"B $\rightarrow$ 2 -1 1N = 100 $\rightarrow$ "CV"

: :

Z $\rightarrow$ 26 -1 25

$$0 \leq N \% 26 <= 25$$

N = 28 $\rightarrow$ "AB"N = 10 $\rightarrow$ binary

2	10	0
2	5	1
2	2	0
2	1	1

0

1010

	N	N-1
26	28 $\rightarrow$	27 % 26 = 1 $\uparrow$ B
26	1 $\rightarrow$	0 % 26 = 0 $\uparrow$ A

AB



	N	N-1		
	26	500	499 % 26	S F
		19	18 % 26	18 S
			0	

0..... 25

pseudocode

ans = ""

while (n > 0) {

n -= 1

rem = n % 26

quo = n / 26

ans = ans + (char)('A' + rem)

n = quo

}

// reverse string

return ans

TC: $O(\log(n)^2)$

If you use string builder

TC: $O(\log(N))$

Doubt session

String Concatenation

$s = \text{"a b c d z "}$
size = n

$s = s + 'a'$ } TC: $O(N)$

Strings are immutable

abcd + 'a'

length of $s = N$

for $i \longrightarrow 0 \text{ to } N-1$ {
 $s = s + 'a'$
}

Heap

abcd

abcda

TC: $O(N^2)$